

GRAPH SIGNAL PROCESSING OF BRAIN CONNECTOME NETWORKS: SPECTRAL ANALYSIS AND GEOMETRIC DEEP LEARNING FOR GLIOBLASTOMA PREDICTION

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ABSTRACT. We develop a graph signal processing (GSP) framework for analyzing atlas-based brain connectomes and training a geometric deep learning model for glioblastoma multiforme (GBM) prediction under an explicit smoothness prior. Regional MRI measurements are represented as signals over a shared 48-node Harvard–Oxford cortical atlas graph, enabling analysis in both the vertex and spectral domains. The graph Fourier transform, Dirichlet energy, spectral centroid, spectral entropy, and multiscale graph wavelets are used to characterize disease-related deviations from healthy brain organization. GBM signals are found to concentrate in lower graph frequencies with reduced spectral entropy compared to healthy cohorts, while multiscale wavelet energy distinguishes cohorts at fine and intermediate scales. A graph neural network classifier trained under a smoothness-regularized objective—combining binary cross-entropy prediction loss with a Dirichlet energy penalty on learned node embeddings—achieves perfect class separation on the reported cohort. The smoothness penalty acts as a graph diffusion operator on learned representations, suppressing graph-inconsistent variation while preserving task-relevant disease contrast. Together, the spectral analysis and regularized learning framework provide a principled, interpretable approach to connectome-aware disease prediction.

1. INTRODUCTION

Brain imaging data are naturally structured: measurements are not independent across space, but are constrained by anatomy, functional relationships, and long-range organization. When regional quantities are aggregated over a cortical atlas, each brain scan may be described as a signal over a set of graph nodes, where nodes correspond to regions of interest (ROIs) and edges encode inter-regional affinity. This representation makes graph signal processing (GSP) a natural mathematical language for studying how disease perturbs spatial organization.

Let

$$G = (V, E, W)$$

denote an undirected weighted graph with node set $V = \{1, \dots, N\}$, edge set E , and symmetric weighted adjacency matrix $W \in \mathbb{R}^{N \times N}$, where $W_{ij} \geq 0$. For each subject s , an imaging-derived regional measurement is represented as a graph signal

$$x^{(s)} \in \mathbb{R}^N,$$

whose i -th entry is the mean signal value in region i . A disease classifier maps the graph signal (and optionally graph structure) to a label

$$y^{(s)} \in \{0, 1\},$$

with $y^{(s)} = 1$ representing GBM and $y^{(s)} = 0$ representing healthy tissue.

The central modeling principle is that biologically meaningful graph signals often exhibit *piecewise smoothness*: neighboring regions tend to have related values unless disrupted by pathology. This motivates regularizing the learned predictive representation using graph Dirichlet energy. The resulting objective balances prediction fidelity against structural smoothness:

$$(1) \quad \mathcal{L}_{\text{total}} = \mathcal{L}_{\text{pred}} + \lambda \mathcal{L}_{\text{smooth}},$$

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where $\mathcal{L}_{\text{pred}}$ is task-specific and $\mathcal{L}_{\text{smooth}}$ is a smoothness penalty derived from graph Dirichlet energy, with regularization weight $\lambda > 0$.

This paper unifies four threads:

- (i) construction of a shared brain graph from anatomical regions,
- (ii) representation of regional measurements as graph signals,
- (iii) spectral and wavelet characterization of disease-related deviations, and
- (iv) supervised learning with a prediction objective regularized by graph smoothness.

The formulation is consistent across all analyses: the spectrum, graph wavelets, and smoothness measures serve both as descriptive tools and as principled regularizers in the supervised learning problem.

2. CONSTRUCTION OF THE BRAIN CONNECTOME GRAPH

2.1. Network Definition. A common atlas provides a shared node set across subjects. The Harvard–Oxford cortical atlas defines $N = 48$ cortical ROIs, giving a fixed graph topology for all subjects. Let the node set be

$$V = \{1, \dots, 48\}.$$

An undirected edge $(i, j) \in E$ is included when regions i and j are considered anatomically or statistically connected in the shared reference graph. The graph is represented by its weighted adjacency matrix W , degree matrix D , and graph Laplacian L :

$$(2) \quad W_{ij} = W_{ji} \geq 0,$$

$$(3) \quad D_{ii} = \sum_{j=1}^N W_{ij},$$

$$(4) \quad L = D - W.$$

Because node degrees may vary, the normalized Laplacian is also useful:

$$(5) \quad L_{\text{sym}} = I - D^{-1/2} W D^{-1/2}.$$

The normalized version permits comparison across regions with different connectivity levels and is particularly convenient for spectral analysis and graph convolutions.

2.2. Atlas and Spatial Registration. The atlas-based construction pipeline is illustrated in Figures 1 and 2. The MNI152 template provides the common reference space, the Harvard–Oxford atlas provides ROI labels, and subject-level MRI and tumor masks are mapped into this space so that node-wise quantities can be computed consistently across the cohort.

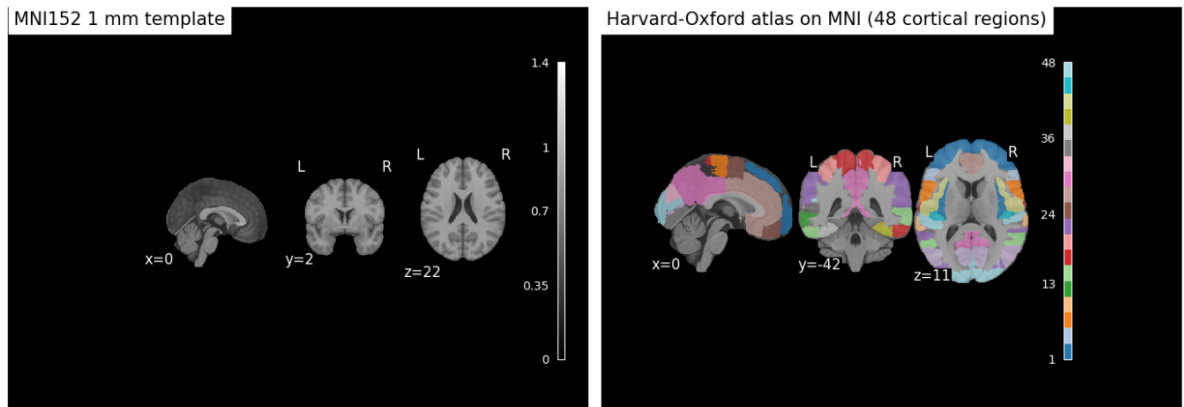


FIGURE 1. Reference space and atlas definition. Left: MNI152 1 mm template. Right: Harvard–Oxford atlas projected onto MNI space with 48 cortical regions. This atlas defines the graph nodes used throughout the analysis.

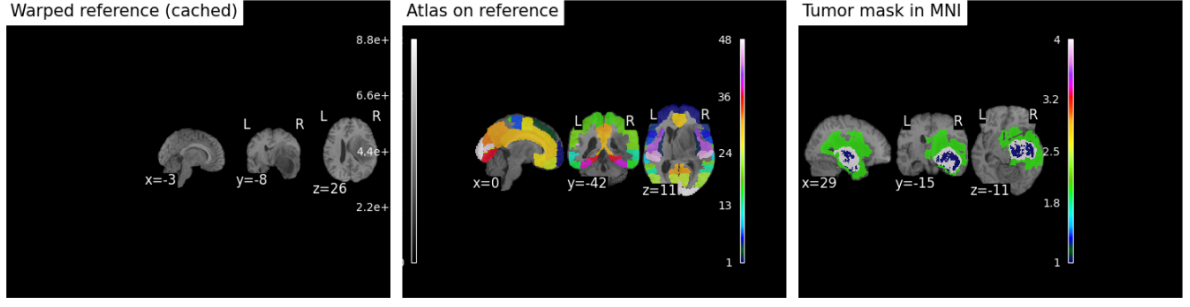


FIGURE 2. Registration and atlas alignment workflow. The reference image, atlas labels, and tumor mask are shown in common space, enabling atlas-wise aggregation of tumor burden and image-derived regional signals.

After registration, each atlas node receives a region-level summary statistic, such as the mean T1 intensity or the fraction of voxels covered by tumor.

2.3. Descriptive Topology of the Shared Connectome. Figure 3 summarizes the degree distribution, clustering coefficient distribution, betweenness centrality distribution, and graph visualizations colored by several node-level attributes.

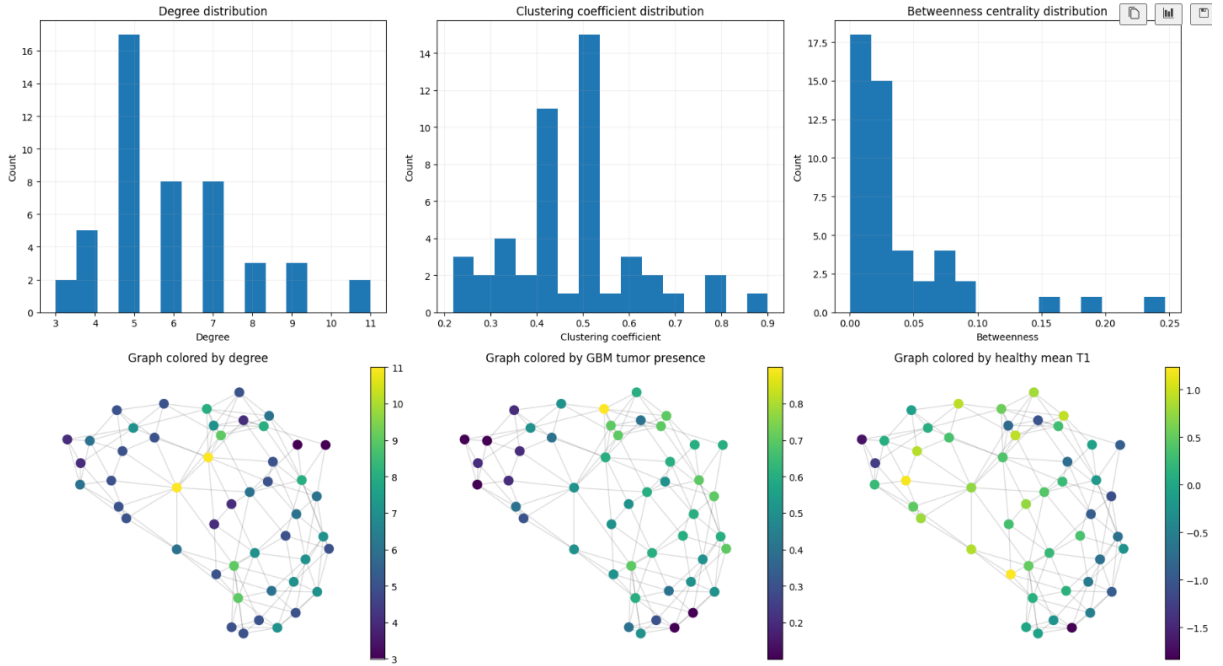


FIGURE 3. Descriptive graph statistics and graph visualizations for the shared 48-node connectome. Top row: distributions of degree, clustering coefficient, and betweenness centrality. Bottom row: the same graph colored by degree, average GBM tumor presence, and average healthy mean T1 signal.

If d_i is the degree of node i , the average degree is

$$(6) \quad \bar{d} = \frac{1}{N} \sum_{i=1}^N d_i.$$

The local clustering coefficient of node i is

$$(7) \quad C_i = \frac{2T_i}{d_i(d_i - 1)},$$

where T_i is the number of triangles through node i . Betweenness centrality quantifies how frequently a node lies on shortest paths:

$$(8) \quad B_i = \sum_{\substack{s \neq i \neq t \\ s < t}} \frac{\sigma_{st}(i)}{\sigma_{st}},$$

where σ_{st} is the number of shortest paths from s to t and $\sigma_{st}(i)$ counts those passing through i . These graph properties matter because a smoothness penalty of the form $f^\top L f$ penalizes differences across edges; nodes with many strong incident edges contribute more strongly to the total regularization.

3. BRAIN MEASUREMENTS AS GRAPH SIGNALS

3.1. Regional Signal Definition. Given a subject-level image $I^{(s)}(\mathbf{r})$ in common space and an atlas region Ω_i , a node-wise regional summary signal is defined as

$$(9) \quad x_i^{(s)} = \frac{1}{\mu(\Omega_i)} \int_{\Omega_i} I^{(s)}(\mathbf{r}) d\mathbf{r}.$$

Stacking across all nodes yields the graph signal $x^{(s)} \in \mathbb{R}^N$.

Two particularly important signals are:

- (a) the mean T1 signal $x_{\text{t1}}^{(s)}$, and
- (b) the tumor burden signal $x_{\text{tumor}}^{(s)}$, where

$$(10) \quad x_{\text{tumor},i}^{(s)} = \frac{1}{|\Omega_i|} \sum_{\mathbf{r} \in \Omega_i} M^{(s)}(\mathbf{r}),$$

with $M^{(s)}(\mathbf{r}) \in \{0, 1\}$ the tumor mask.

The tumor burden by atlas node for a representative case is shown in Figure 4, revealing spatially nonuniform disease distribution over the connectome.

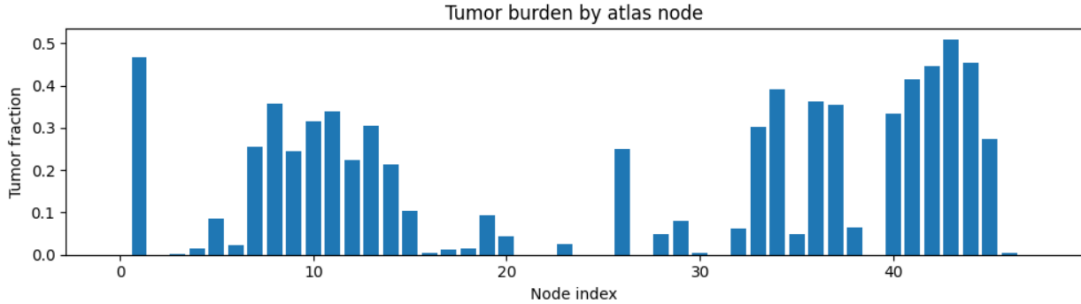


FIGURE 4. Tumor burden by atlas node for a representative GBM subject. The node-wise tumor fraction defines a graph signal that can be analyzed spectrally and spatially.

3.2. Cohort Mean Signals and Disease-Related Spatial Contrast. Let $\mathcal{S}_{\text{GBM}}$ and $\mathcal{S}_{\text{H}}$ denote the GBM and healthy cohorts. Their mean signals are

$$(11) \quad \bar{x}_{\text{GBM}} = \frac{1}{|\mathcal{S}_{\text{GBM}}|} \sum_{s \in \mathcal{S}_{\text{GBM}}} x^{(s)},$$

$$(12) \quad \bar{x}_{\text{H}} = \frac{1}{|\mathcal{S}_{\text{H}}|} \sum_{s \in \mathcal{S}_{\text{H}}} x^{(s)}.$$

A natural disease contrast is then

$$(13) \quad \Delta x = \bar{x}_{\text{GBM}} - \bar{x}_{\text{H}}.$$

Figure 5 visualizes the mean GBM signal, mean healthy signal, and the cohort-level difference over the shared connectome. The disease effect is not spatially uniform; instead it is concentrated over subsets of regions, motivating graph-based multiscale analysis.

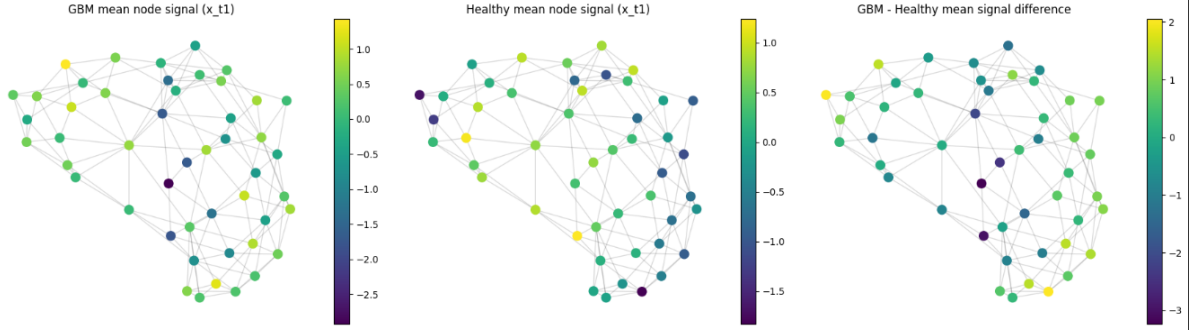


FIGURE 5. Cohort mean graph signals and their difference. Left: mean GBM node signal. Middle: mean healthy node signal. Right: GBM minus healthy mean signal. Disease-related deviations are spatially structured over the graph rather than randomly distributed.

Figure 6 shows the average tumor presence over the shared graph, providing a biologically interpretable disease burden map.

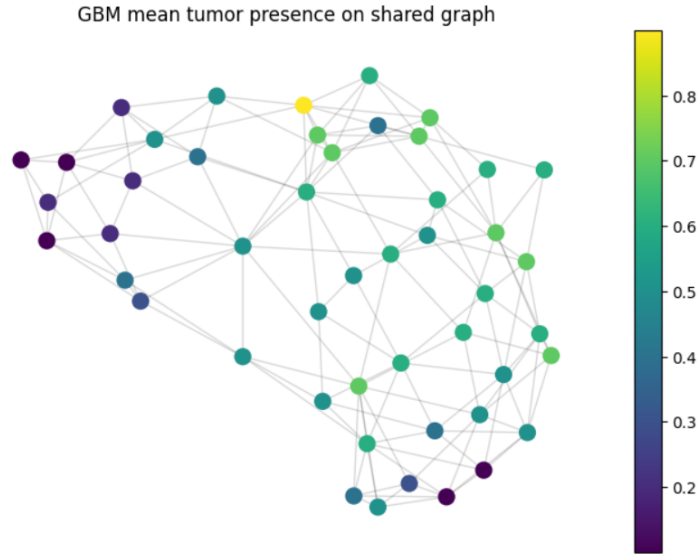


FIGURE 6. Average GBM tumor presence projected onto the shared graph. This cohort-level map indicates which atlas nodes are most frequently involved by tumor across GBM subjects.

4. SPECTRAL GRAPH ANALYSIS OF BRAIN SIGNALS

4.1. Graph Fourier Transform. The graph Laplacian L is real symmetric and admits the eigendecomposition

$$(14) \quad L = U\Lambda U^\top,$$

where

$$U = [u_1, \dots, u_N], \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_N),$$

with $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N$. The graph Fourier transform (GFT) of a signal $x \in \mathbb{R}^N$ is defined by

$$(15) \quad \hat{x} = U^\top x,$$

with inverse transform

$$(16) \quad x = U\hat{x} = \sum_{k=1}^N \hat{x}_k u_k.$$

Low graph frequencies correspond to Laplacian eigenvectors with small eigenvalues and vary slowly over strongly connected regions, whereas high graph frequencies oscillate more sharply across neighboring nodes.

4.2. Dirichlet Energy and Spectral Smoothness. A central quantity is the graph Dirichlet energy:

$$(17) \quad \mathcal{E}(x) = x^\top Lx = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N W_{ij} (x_i - x_j)^2.$$

Equation (17) gives two equivalent interpretations:

- (1) in the vertex domain, it penalizes edge-wise disagreement;
- (2) in the spectral domain, it weights energy by graph frequency.

Indeed, using $L = U\Lambda U^\top$,

$$(18) \quad \mathcal{E}(x) = x^\top U\Lambda U^\top x = \hat{x}^\top \Lambda \hat{x} = \sum_{k=1}^N \lambda_k |\hat{x}_k|^2.$$

Thus a signal is smooth if most of its spectral mass lies at low eigenvalues.

4.3. Spectral Biomarkers. Several scalar spectral descriptors summarize graph signals compactly.

Spectral centroid. The spectral centroid quantifies where the signal is centered in graph frequency:

$$(19) \quad C_{\text{spec}}(x) = \frac{\sum_{k=1}^N \lambda_k |\hat{x}_k|^2}{\sum_{k=1}^N |\hat{x}_k|^2}.$$

This is the energy-weighted average graph frequency. Higher values imply a shift toward less smooth components.

Spectral entropy. Define normalized spectral weights $p_k = |\hat{x}_k|^2 / \sum_{\ell} |\hat{x}_\ell|^2$. Then the spectral entropy is

$$(20) \quad H_{\text{spec}}(x) = - \sum_{k=1}^N p_k \log p_k.$$

Higher entropy indicates a more diffuse spectral distribution.

Low- and high-frequency energy. Let $\mathcal{K}_{\text{low}}$ and $\mathcal{K}_{\text{high}}$ denote index sets of low and high graph frequencies. Then

$$(21) \quad E_{\text{low}}(x) = \sum_{k \in \mathcal{K}_{\text{low}}} |\hat{x}_k|^2, \quad E_{\text{high}}(x) = \sum_{k \in \mathcal{K}_{\text{high}}} |\hat{x}_k|^2.$$

Larger E_{high} indicates sharper spatial variation and reduced smoothness.

Figure 7 compares the mean spectra of the GBM and healthy cohorts. The healthy cohort shows stronger content across a broader range of graph frequencies, while the GBM cohort exhibits a different distribution of spectral power, especially in lower-to-mid frequencies.

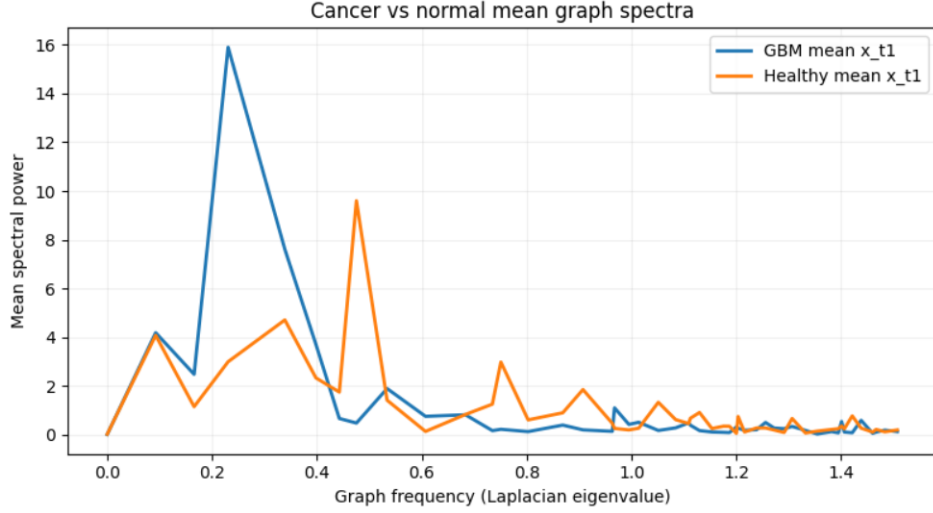


FIGURE 7. Mean graph spectra for GBM and healthy cohorts. The spectra are plotted as spectral power versus graph frequency (Laplacian eigenvalue), highlighting disease-related redistribution of energy across graph harmonics.

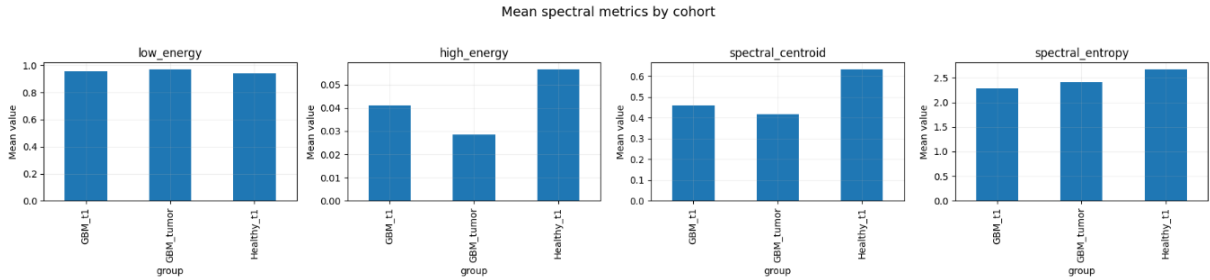


FIGURE 8. Mean spectral metrics by cohort: low-frequency energy, high-frequency energy, spectral centroid, and spectral entropy. These compact descriptors summarize the organization of graph signal energy across the spectrum.

The distributional comparisons in Figure 8 confirm that the healthy cohort tends to exhibit larger high-frequency energy, larger centroid, and larger entropy than the GBM tumor signal. Table 1 summarizes the interpretation of these spectral biomarkers.

TABLE 1. Spectral biomarkers and their interpretation in the connectome graph setting.

Metric	Formula	Interpretation
Low-frequency energy	$E_{\text{low}} = \sum_{k \in \mathcal{K}_{\text{low}}} \hat{x}_k ^2$	Measures signal power concentrated in smooth graph harmonics. Larger values imply stronger global smoothness.
High-frequency energy	$E_{\text{high}} = \sum_{k \in \mathcal{K}_{\text{high}}} \hat{x}_k ^2$	Measures signal power concentrated in rapidly varying graph harmonics. Larger values imply sharper regional contrast.
Spectral centroid	$C_{\text{spec}} = \frac{\sum_k \lambda_k \hat{x}_k ^2}{\sum_k \hat{x}_k ^2}$	Energy-weighted average graph frequency. Higher values indicate the spectrum is shifted toward less smooth components.
Spectral entropy	$H_{\text{spec}} = -\sum_k p_k \log p_k$	Quantifies how diffuse the spectrum is. Higher values indicate more distributed energy across modes.

5. GRAPH WAVELET ANALYSIS

Spectral descriptors summarize overall distribution of graph frequency content but do not explicitly preserve localization in both vertex and scale domains. Graph wavelets address this by defining multiscale filters on the Laplacian spectrum.

5.1. Spectral Graph Wavelets. Let $g(\cdot)$ be a spectral kernel. A graph wavelet operator at scale $\tau > 0$ is

$$(22) \quad T_g^{(\tau)} = g(\tau L) = U g(\tau \Lambda) U^\top,$$

where $g(\tau \Lambda) = \text{diag}(g(\tau \lambda_1), \dots, g(\tau \lambda_N))$. Applying this operator to a graph signal x gives the scale-dependent wavelet coefficients

$$(23) \quad w^{(\tau)} = T_g^{(\tau)} x.$$

For heat-kernel-style diffusion wavelets one may use $g(\tau \lambda) = e^{-\tau \lambda}$, where small τ emphasizes finer detail and large τ emphasizes coarser smoothing. The scale-dependent wavelet energy

$$(24) \quad E_{\text{wav}}(\tau; x) = \|w^{(\tau)}\|_2^2 = \sum_{k=1}^N |g(\tau \lambda_k)|^2 |\hat{x}_k|^2$$

provides a useful bridge between descriptive multiscale analysis and graph-regularized learning.

5.2. Multiscale Disease Separation. Figure 9 shows multiscale graph wavelet energy at $\tau = 1, 5, 10, 20$. Healthy T1 signals exhibit markedly larger wavelet energy at smaller and intermediate scales, while the GBM tumor signal remains comparatively small and compressed. Wavelet energy is especially informative because signals with high fine-scale wavelet energy necessarily retain more localized irregularity on the graph, which is closely related to larger Dirichlet energy and therefore to a larger smoothness penalty in the learning objective.

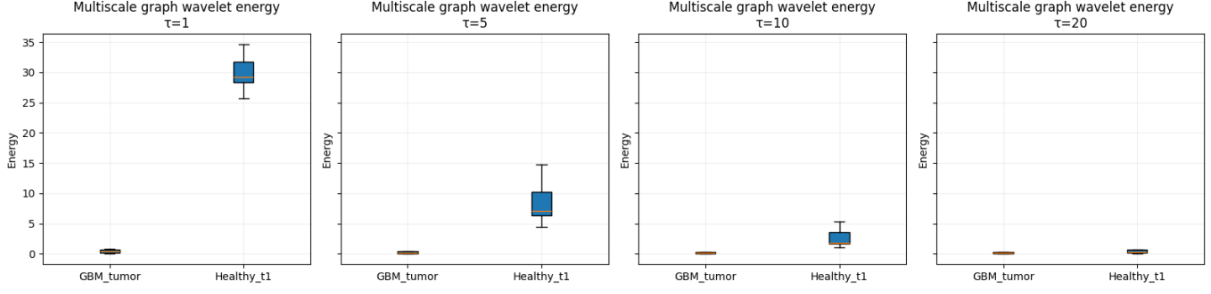


FIGURE 9. Multiscale graph wavelet energy across cohorts at scales $\tau = 1, 5, 10, 20$. The strongest group differences occur at smaller scales, where healthy T1 signals display substantially larger wavelet energy than GBM tumor signals.

6. DIFFUSION DYNAMICS AND DIRICHLET SMOOTHNESS

6.1. Heat Diffusion Model. A standard physical interpretation of graph smoothness is diffusion. Let $f(t) \in \mathbb{R}^N$ be a time-dependent graph signal satisfying

$$(25) \quad \frac{df(t)}{dt} = -Lf(t), \quad f(0) = f_0.$$

The solution is

$$(26) \quad f(t) = e^{-tL} f_0 = U e^{-t\Lambda} U^\top f_0.$$

Because each spectral component decays as $e^{-t\lambda_k}$, high-frequency components vanish more rapidly than low-frequency ones. The derivative of the squared norm under heat diffusion satisfies

$$(27) \quad \frac{d}{dt} \|f(t)\|_2^2 = -2f(t)^\top L f(t) = -2\mathcal{E}(f(t)),$$

so Dirichlet energy measures the instantaneous rate at which signal energy is dissipated by diffusion. Signals with large Dirichlet energy contain sharp edge-wise discrepancies and smooth quickly; signals with small Dirichlet energy are already smooth and diffuse slowly.

6.2. Interpretation for Brain Signals. The smoothness assumption does *not* mean that disease effects are spatially uniform. Rather, it imposes that the *learned predictive representation* should avoid gratuitous node-to-node oscillation not supported by the graph structure. The objective is not to maximize smoothness at all costs but to minimize

$$\mathcal{L}_{\text{pred}} + \lambda \mathcal{L}_{\text{smooth}},$$

so that the retained variation is the variation needed for discrimination.

7. GEOMETRIC DEEP LEARNING FOR CONNECTOME PREDICTION

7.1. Model Overview. Given a graph $G = (V, E, W)$ and subject signal $x^{(s)}$, the goal is to predict whether the subject belongs to the GBM or healthy cohort. A graph neural network (GNN) maps node features to a graph-level representation and then to a class probability:

$$(28) \quad \hat{y}^{(s)} = \sigma(\phi_{\text{readout}}(H^{(L,s)})),$$

where $H^{(L,s)} \in \mathbb{R}^{N \times F_L}$ is the node embedding matrix after L layers, ϕ_{readout} is a graph-level pooling function, and σ is the logistic sigmoid.

7.2. Message Passing and Graph Convolution. A standard message passing layer is

$$(29) \quad h_i^{(\ell+1)} = \psi^{(\ell)} \left(h_i^{(\ell)}, \bigoplus_{j \in \mathcal{N}(i)} \phi^{(\ell)}(h_i^{(\ell)}, h_j^{(\ell)}, e_{ij}) \right),$$

where $\oplus$ is a permutation-invariant aggregation. For a graph convolution using the normalized adjacency,

$$(30) \quad H^{(\ell+1)} = \rho \left(\tilde{D}^{-1/2} \tilde{W} \tilde{D}^{-1/2} H^{(\ell)} \Theta^{(\ell)} \right),$$

where $\tilde{W} = W + I$, $\tilde{D}$ is its degree matrix, $\Theta^{(\ell)}$ are learnable weights, and ρ is a pointwise nonlinearity. Equation (30) shows explicitly how the graph topology constrains feature mixing across neighboring regions.

7.3. Graph-Level Readout. The graph-level embedding is formed by global mean pooling,

$$(31) \quad z^{(s)} = \frac{1}{N} \sum_{i=1}^N h_i^{(L,s)},$$

and the predicted probability is

$$(32) \quad \hat{p}^{(s)} = \sigma(w^\top z^{(s)} + b) = \frac{1}{1 + e^{-\eta^{(s)}}}.$$

7.4. Training Objective. The task-specific prediction loss is binary cross-entropy:

$$(33) \quad \mathcal{L}_{\text{pred}} = -\frac{1}{M} \sum_{s=1}^M \left[y^{(s)} \log \hat{p}^{(s)} + (1 - y^{(s)}) \log (1 - \hat{p}^{(s)}) \right].$$

The smoothness penalty uses a matrix-valued generalization of graph Dirichlet energy over learned embeddings $H^{(L,s)}$:

$$(34) \quad \mathcal{L}_{\text{smooth}} = \frac{1}{M} \sum_{s=1}^M \text{Tr} \left((H^{(L,s)})^\top L H^{(L,s)} \right) = \frac{1}{2M} \sum_{s=1}^M \sum_{i,j} W_{ij} \|h_i^{(L,s)} - h_j^{(L,s)}\|_2^2.$$

Thus neighboring nodes with strong graph affinity are encouraged to have similar final embeddings unless task-specific gradients force them apart. The gradient of the smoothness term with respect to $H^{(L,s)}$ is

$$(35) \quad \frac{\partial}{\partial H^{(L,s)}} \text{Tr} \left((H^{(L,s)})^\top L H^{(L,s)} \right) = 2LH^{(L,s)},$$

so the smoothness gradient acts as a graph diffusion operator on learned node embeddings.

Table 2 summarizes the components of the training objective.

TABLE 2. Training objective components and interpretations.

Component	Formula	Purpose
Prediction loss	$\mathcal{L}_{\text{pred}} = -\frac{1}{M} \sum_s [y^{(s)} \log \hat{p}^{(s)} + (1 - y^{(s)}) \log (1 - \hat{p}^{(s)})]$	Encourages accurate subject-level disease classification.
Smoothness penalty	$\mathcal{L}_{\text{smooth}} = \frac{1}{M} \sum_s \text{Tr}((H^{(L,s)})^\top L H^{(L,s)})$	Penalizes edge-wise discrepancies in learned node embeddings.
Total objective	$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{pred}} + \lambda \mathcal{L}_{\text{smooth}}$	Balances classification performance against graph-structured regularity.

8. RESULTS AND DISCUSSION

8.1. Geometric Deep Learning Results. The geometric deep learning workflow and training curves are shown in Figure 10. The training curves reveal three important patterns:

- (a) the total loss decreases smoothly on both train and test sets;
- (b) classification accuracy improves to perfect separation on the reported split;
- (c) the smoothness penalty increases as training proceeds, while the prediction loss decreases.

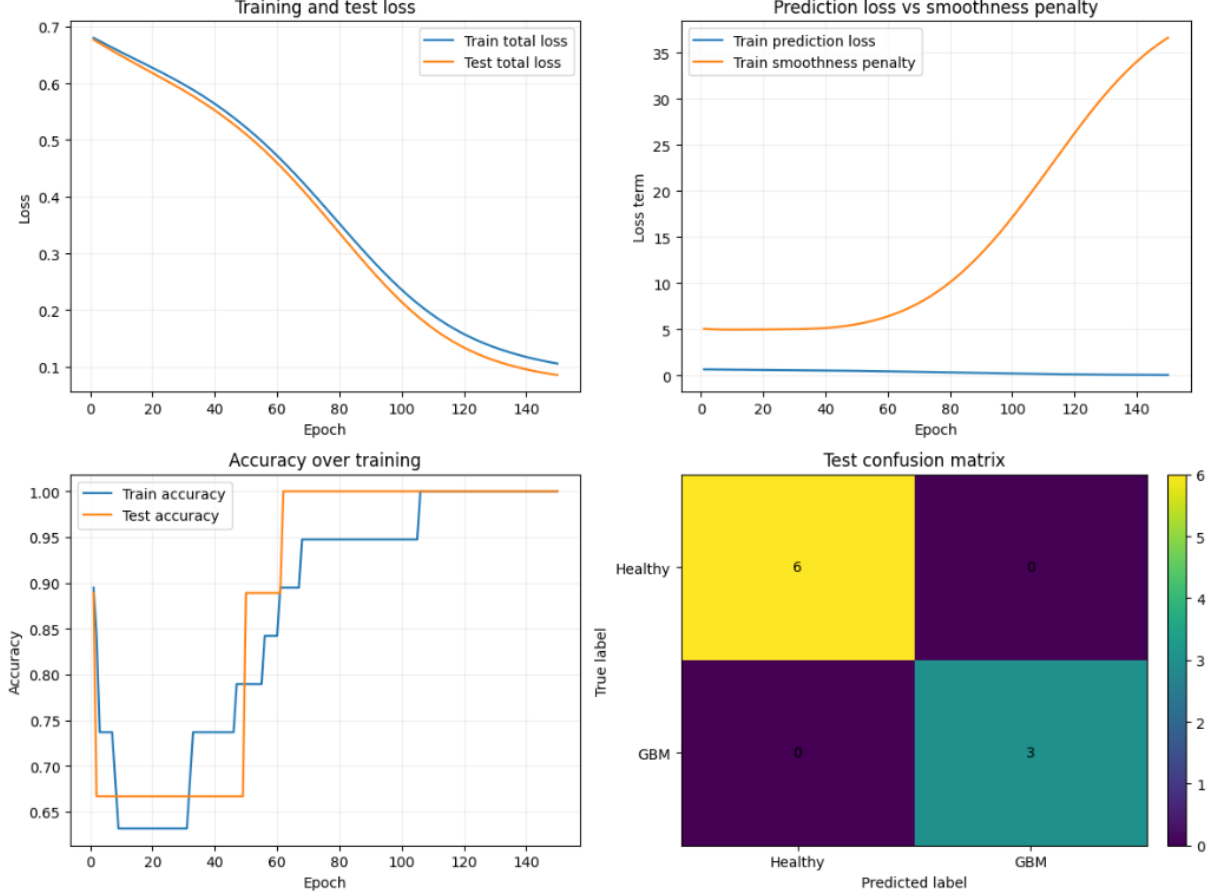


FIGURE 10. Geometric deep learning results. Top-left: training and test total loss curves. Top-right: evolution of the task-specific prediction loss and the smoothness penalty. Bottom-left: training and test accuracy. Bottom-right: test confusion matrix showing perfect separation on the reported split.

The third pattern is particularly informative: to improve discrimination, the model learns embeddings that are less smooth than at initialization. This is not contradictory—the objective minimizes the *sum* of task loss and smoothness penalty, so the optimum may legitimately move toward a higher smoothness term when that increase is more than compensated by a larger reduction in $\mathcal{L}_{\text{pred}}$. The regularizer does not prohibit task-relevant graph variation; it discourages gratuitous variation.

The test confusion matrix shows 6/6 healthy and 3/3 GBM examples correctly classified on the reported split. The PCA projection of learned subject embeddings (Figure 11) further demonstrates clear class separation.

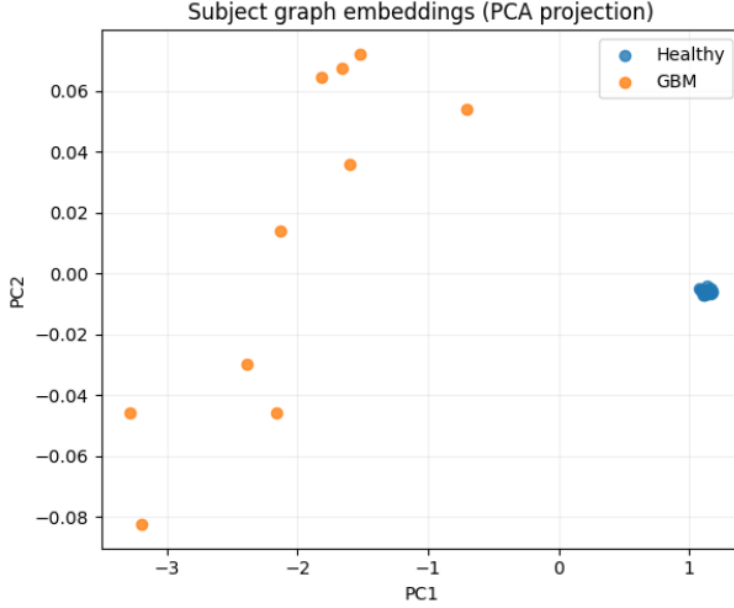


FIGURE 11. PCA projection of learned subject embeddings. Healthy subjects form a compact cluster, while GBM subjects occupy a well-separated region of the embedding space.

8.2. Spectral Biomarker Summary. The descriptive analyses confirm that disease and healthy cohorts differ in how signal energy is distributed over graph frequencies and graph scales. The key findings are:

- GBM tumor signals are more concentrated in lower graph frequencies;
- healthy T1 signals exhibit higher spectral centroid, higher entropy, and higher high-frequency energy;
- multiscale wavelet energy is largest in healthy signals, especially at smaller scales;
- disease effects are localized over specific atlas nodes and subnetworks rather than uniformly distributed.

These observations justify using graph-structured regularization in a classifier: the graph carries biologically meaningful spatial organization, and the spectrum provides an interpretable basis for quantifying disease-related departures from typical organization.

8.3. Stability and Robustness. The methodology combines a fixed shared graph with subject-varying signals, offering several stability advantages:

- (1) **Common node semantics:** every node corresponds to the same atlas region across subjects;
- (2) **Shared spectral basis:** the eigenvectors of the common graph provide a single reference frame for all GFT analyses;
- (3) **Regularized learning:** the Dirichlet penalty reduces the risk of overfitting to noisy subject-specific node fluctuations.

If the observed signal is $x = x^* + \varepsilon$ where ε is noise, then

$$(36) \quad \mathcal{E}(x) = \mathcal{E}(x^*) + 2(x^*)^\top L\varepsilon + \mathcal{E}(\varepsilon),$$

and noise concentrated at high graph frequencies inflates the Dirichlet energy because $\mathcal{E}(\varepsilon) = \sum_k \lambda_k |\hat{\varepsilon}_k|^2$. Thus the smoothness penalty preferentially suppresses high-frequency noise, giving it a denoising interpretation in addition to its role as a structural prior.

9. CONCLUSION

This paper establishes a full graph signal processing pipeline for atlas-based brain connectome analysis of glioblastoma multiforme. The main contributions are:

- (1) a shared 48-node Harvard–Oxford atlas graph representation for all subjects,
- (2) graph signal definitions for T1 intensity and tumor burden,
- (3) spectral biomarkers including high-frequency energy, spectral centroid, and spectral entropy,
- (4) multiscale graph wavelet analysis of disease-related signals,
- (5) a diffusion-based interpretation of smoothness via Dirichlet energy, and
- (6) a geometric deep learning classifier trained under a smoothness-regularized objective.

The central conclusion is that connectome-aware analysis is valuable both descriptively and predictively. Descriptively, the graph Fourier basis and graph wavelets reveal where disease-related information is located across graph frequencies and scales. Predictively, the same graph supports the smoothness-regularized learning objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{pred}} + \lambda \mathcal{L}_{\text{smooth}},$$

which is principled because it reflects the assumption that neighboring brain regions should generally have related latent representations, except where the classification task benefits from preserving edge-wise contrast. The results support the view that disease-related brain patterns are best understood as graph signals over a biologically meaningful network, and that the combination of spectral analysis, graph wavelets, and smoothness-regularized geometric learning provides a coherent framework for both mechanistic interpretation and accurate prediction.

TABLE 3. Compact notation summary.

Symbol	Meaning
$G = (V, E, W)$	Weighted undirected connectome graph with node set V , edge set E , and adjacency matrix W .
N	Number of graph nodes (here $N = 48$).
D	Degree matrix, $D_{ii} = \sum_j W_{ij}$.
L	Combinatorial graph Laplacian, $L = D - W$.
L_{sym}	Symmetric normalized Laplacian, $I - D^{-1/2}WD^{-1/2}$.
$x^{(s)}$	Graph signal for subject s .
$\hat{x}$	Graph Fourier transform of x , $\hat{x} = U^\top x$.
U, Λ	Laplacian eigenvectors and eigenvalues, $L = U\Lambda U^\top$.
$\mathcal{E}(x)$	Dirichlet energy, $x^\top Lx$.
E_{high}	High-frequency spectral energy.
C_{spec}	Spectral centroid.
H_{spec}	Spectral entropy.
$H^{(L,s)}$	Final-layer node embedding matrix for subject s .
$\mathcal{L}_{\text{pred}}$	Task-specific prediction loss (binary cross-entropy).
$\mathcal{L}_{\text{smooth}}$	Smoothness penalty based on Dirichlet energy of learned embeddings.
λ	Regularization strength balancing prediction and smoothness.

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